Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In compounds of type ECl_3, where $E = B, P$, As or Bi, the angles $Cl - E - Cl$ for different E are in the order.

(A) $B > P = As = Bi$

(B) $B > P > As > Bi$

(C) $B < P = As = Bi$

(D) $B < P < As < Bi$

Q.2

In which of the following species the interatomic bond angle is $109°28'$?

(A) NH_4^+, BF_4^-

(B) NH_4^+, BF_3

(C) NH_3, BF_4

(D) NH_2^-, BF_3.

Q.3

The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively

(A) sp^3 and sp^2

(B) sp^2 and sp^3

(C) sp^2 and sp^2

(D) sp^3 and sp^3

Q.4

The shape of IF_6^- is :

(A) Trigonally distorted octahedron

(B) Pyramidal

(C) Octahedral

(D) Square antiprism

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Q.5

The bond angle between two hybrid orbitals is 105° Calculate the percentage of s -character of hybrid orbital.

(A) 21.55

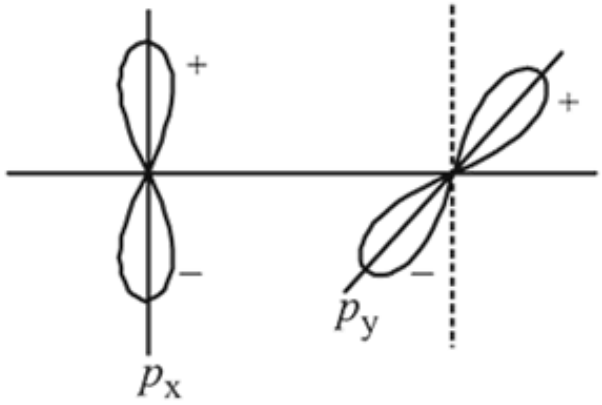
(B) 20.44

(C) 19.45

(D) 21.44

Q.6

Main axis of a diatomic molecule is Z. AO's p_x and p_y overlap to form which of the following orbitals?



(A) π -MO

(B) $\sigma - \mathrm{MO}$

(C) $\delta - \mathrm{MO}$

(D) No bond will form

Q.7

The group having triangular planar structures is:

(A) $\mathrm{BF_3}, \mathrm{NF_3}, \mathrm{CO_3^{2-}}$

(B) $\mathrm{CO_3^{2-}}, \mathrm{NO_3^-}, \mathrm{SO_3}$

(C) $\mathrm{NH_3}, \mathrm{SO_3}, \mathrm{CO_3^{2-}}$

(D) $\mathrm{NCl_3}, \mathrm{BCl_3}, \mathrm{SO_3}$

Q.8

The compound $\mathrm{MX_4}$ is tetrahedral. The number of $\angle \mathrm{XMX}$ formed in the compound are

(A) three

(B) four

(C) five

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(D) six

Q.9

In a regular octahedral molecule, MX_6 the number of $X-M-X$ bonds at 180° is

(A) three

(B) two

(C) six

(D) four

Q.10

Which of the following molecular orbitals has two nodal planes?

(A) $\sigma 2s$

(B) $\pi 2p_y$

(C) $\pi^* 2p_y$

(D) $\sigma^* 2p_x$

Q.11

In the reaction $2PCl_5 \rightleftharpoons PCl_4^+ + PCl_6^-$, the change in hybridisation is from

(A) sp^3d to sp^3 and sp^3d^2

(B) sp^3d to sp^2 and sp^3

(C) sp^3d to sp^3d^2 and sp^3d^3

(D) sp^3d^2 to sp^3 and sp^3d

Q.12

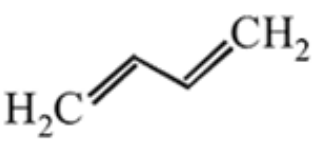
Which of the following represents the given mode of hybridisation $sp^2-sp^2-sp-sp$ from left to right ?

(A) $H_2C=CH-C\equiv N$

(B) $HC\equiv C-C\equiv CH$

(C) $H_2C=C=C=CH_2$

(D)



Q.13

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The group of molecules having identical shape is :

(A) PCl_5, IF_5, XeO_2F_2

(B) BF_3, PCl_3, XeO_3

(C) SF_4, XeF_4, CCl_4

(D) $ClF_3, XeOF_2, XeF_3^+$

Q.14

The maximum number of 90° angles between bond pair-bond pair of electrons is observed in

(A) dsp^2 hybridization

(B) sp^3d hybridization

(C) dsp^3 hybridization

(D) sp^3d^2 hybridization

Q.15

In which of the following sets, all the given species are isostructural?

(A) CO_2, NO_2, ClO_2, SiO_2

(B) $PCl_3, AlCl_3, BCl_3, SbCl_3$

(C) BF_3, NF_3, PF_3, AlF_3

(D) $BF_4^-, CCl_4, NH_4^+, PCl_4^+$

Q.16

Which of the following has the square planar structure?

(A) XeF_4

(B) NH_4^+

(C) BF_4^-

(D) CCl_4

Q.17

Which of the following statement is correct?

(A) H_2O and OF_2 both are linear in shape

(B) H_2O is linear whereas OF_2 is bent molecule

(C) both H_2O and OF_2 have a bent structure

(D) H_2O is bent whereas OF_2 has a linear structure

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Q.18

Which one of the following molecules will form a linear polymeric structure due to H-bonding?

(A) HCl

(B) HF

(C) H_2O

(D) NH_3

Q.19

On hybridization of one s and one p-orbital we get

(A) two mutually perpendicular orbitals

(B) two orbitals at 180°

(C) four orbitals directed tetrahedrally

(D) three orbitals in a plane

Q.20

Hybridization of each carbon in C_3O_2 is

(A) sp

(B) sp^2

(C) sp^3

(D) $sp^2 d$

Answers & Solutions

Q.1 Answer:

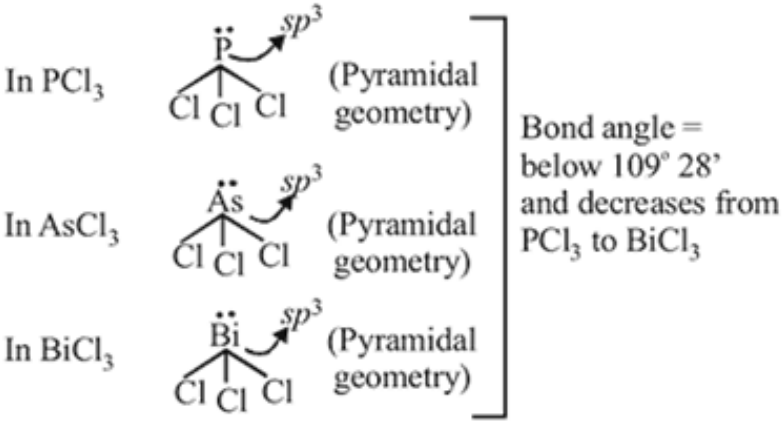
$B > P > As > Bi$

Solution:

$BiCl_3$: $Cl-Bi\begin{matrix}Cl\\Cl\end{matrix}$; sp^2 - Hybridisation

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(Trigonal planar geometry); Bond angle $= 120^{\circ}$



In these, order of bond angle:

$BCl_3 > PCl_3 > AsCl_3 > BiCl_3$

Q.2 Answer:

NH_4^+, BF_4^-

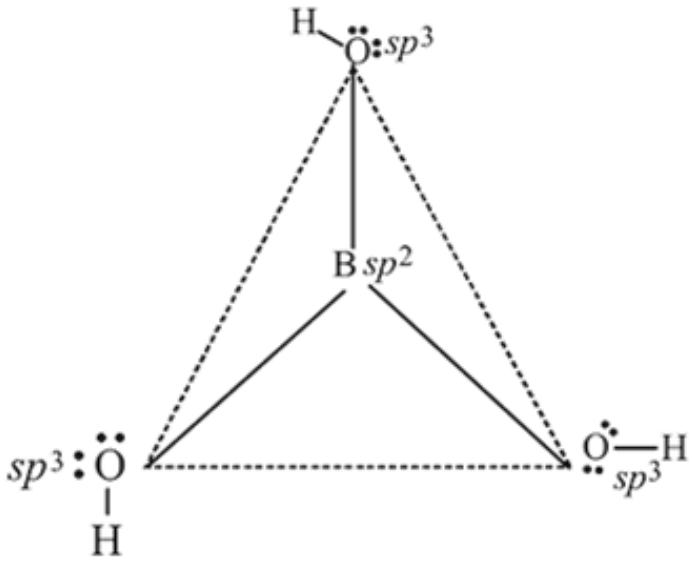
Solution:

In NH_4^+ **and** BF_4^- **the hybridisation is** sp^3 **and the bond angle is almost** $109^{\circ}28'$**.**

Q.3 Answer:

sp^2 and sp^3

Solution:



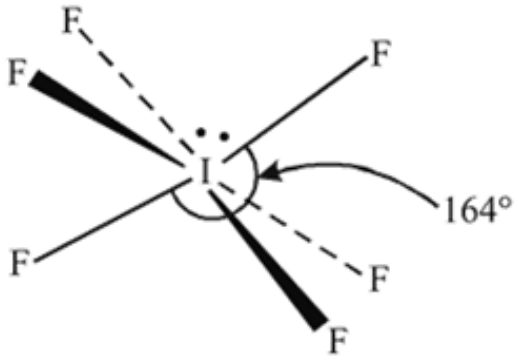
Q.4 Answer:

Trigonally distorted octahedron

Solution:

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The structure of IF_6^- is distorted octahedral This is due to presence of a "weak" lone pair.



Q.5 Answer:

21.44

Solution:

s -character $\propto$ bond angle

For $25\% s$ character (as in sp^3 hybrid orbital), bond angle is 109.5°, for $33.3\% s$ character (as in sp^2 hybrid orbital), bond angle is 120° and for $50\% s$ character (as in sp hybrid orbital), bond angle is 180°. Similarly, when the bond angle decreases below 109.5°, the s -character will decrease accordingly. Decrease in angle $= 120^\circ - 109.5^\circ = 10.5^\circ$

Decrease in s -character $= 33.3 - 25 = 8.3$

Actual decrease in bond angle $= 109.5^\circ - 105^\circ = 4.5^\circ$

Expected decrease in s -character

$= \frac{8.3}{10.5} \times 4.5 = 3.56\%$

Thus, the s -character should decrease by about 3.56%

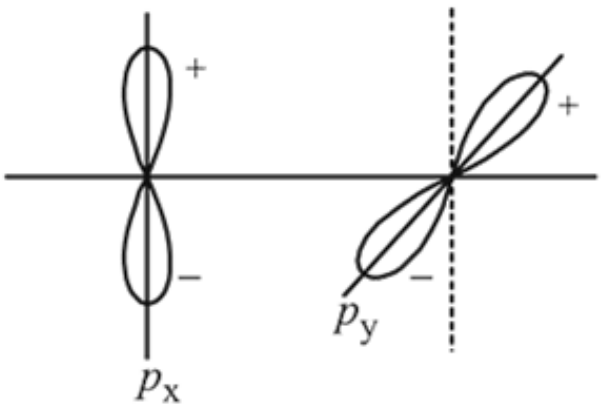
i.e.,s -character $= 25 - 3.56 = 21.44\%$

Q.6 Answer:

No bond will form

Solution:

p_x and p_y orbitals do not have proper orientation to overlap and hence no bond is formed.



Q.7 Answer:

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CO_3^{2-}, NO_3^-, SO_3

Solution:

GroupHybridization Shape

(1)

BF_3 sp^2 Triangular Planar (TP)

NF_3 sp^3 Tetrahedral (T)

CO_3^{2-} sp^2 T.P.

(2) CO_3^{2-} sp^2 **T.P.**

NO_3^- sp^2 T.P. } (All have same hybridization)

SO_2 sp^2 **T.P.**

(3)

NH_3 sp^2 *TP*

SO_3 sp^2 T.P.

CO_3^{2-} sp^2 **T.P.**

(4)

NCl_3 sp^3 T

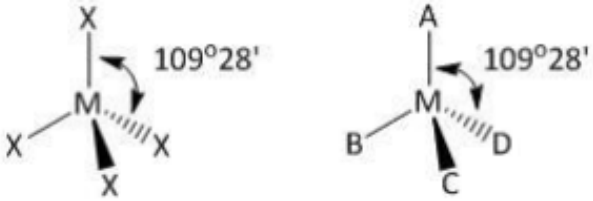
BCl_3 sp^2 T.P.

SO_3 sp^2 T.P.

Q.8 Answer:

six

Solution:



In the tetrahedral model the top X atom is labeled as A, the bottom three are labeled as B, C and D. The atom C is above the plane and the atom D is inside the plane.

Thus the bond angles in MX_4 are written as

$\angle AMB$

$\angle AMC$

$\angle AMD$

$\angle BMC$

$\angle BMD$

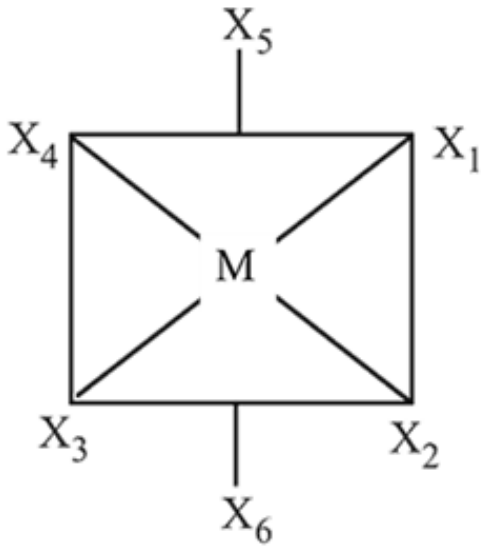
$\angle CMD$

Q.9 Answer:

three

Solution:

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Thus here bond angles between

$X_4 - M - X_2 = 180^\circ$

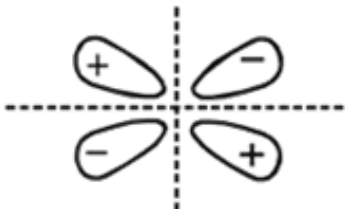
$X_1 - M - X_3 = 180^\circ$

$X_5 - M - X_6 = 180^\circ$

Q.10 Answer:

$\pi^* 2p_y$

Solution:



It has two nodal planes. It is $\pi * 2p_y$

Q.11 Answer:

sp^3d to sp^3 and sp^3d^2

Solution:

$2PCl_5 \rightleftharpoons PCl_4^+ + PCl_6^-$

$\uparrow sp^3d \quad \uparrow sp^3 \uparrow sp^3d^2$

Q.12 Answer:

$H_2C = CH - C \equiv N$

Solution:

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$CH_2 = CH - C \equiv N$

1 2 3 4

3σ bonds $(sp^2$ hybridisation);

2σ bonds $(sp$ -hybridisation)

$C_1 = 3\sigma$ bonds, $C_2 = 3\sigma$ bonds

$C_3 = 2\sigma$ bonds

Q.13 Answer:

$ClF_3, XeOF_2, XeF_3^+$

Solution:

$ClF_3 \longrightarrow$ Hybridisation

$= 3 + \frac{1}{2}[7 - 3] = 5\ (sp^3d)$

$XeOF_2 \longrightarrow$ Hybridisation

$= 3 + \frac{1}{2}[8 - 4] = 5\ (sp^3d)$

$XeF_3^+ \longrightarrow$ Hybridisation

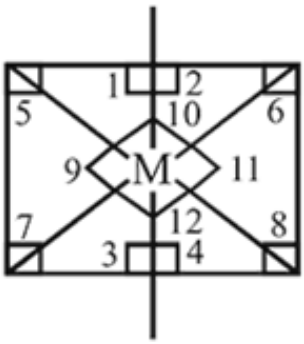
$= 3 + \frac{1}{2}[8 - 3 - 1] = 5\ (sp^3d)$

All molecules have sp^3d hybridization and 2 lone pairs. Hence all have identical (T-shape).

Q.14 Answer:

sp^3d^2 hybridization

Solution:



sp^3d^2 hybridisation

Number of 90° angle between bonds $= 12$

Q.15 Answer:

$BF_4^-, CCl_4, NH_4^+, PCl_4^+$

Solution:

All have tetrahedral structure.

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Q.16 Answer:

XeF_4

Solution:

XeF_4 has square planar structure, while NH_4^+, BF_4^- and CCl_4 have tetrahedral structure.

Q.17 Answer:

both H_2O and OF_2 have a bent structure

Solution:



H_2O and OF_2 have a bent structure.

Q.18 Answer:

HF

Solution:



Q.19 Answer:

two orbitals at 180°

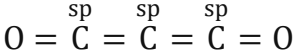
Solution:

Bond angle between two *sp* -hybrid orbitals is 180°.

Q.20 Answer:

sp

Solution:



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